

YOLOv11-Based Apple Leaf Disease Detection: A Comprehensive Literature Review

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Abstract—Apple leaf disease detection has emerged as a critical component of modern horticultural management, driven by the need for rapid, accurate identification of symptoms that threaten yield and fruit quality. This literature review synthesizes recent advances in deep-learning-based vision systems, focusing on the adoption of the YOLOv11 architecture and its lightweight variants for real-time monitoring in orchard environments. The analysis examines methodological trends, including architectural refinements that reduce computational load while preserving fine-grained feature representation, and extensions that enable multi-class detection of co-infecting pathogens. By evaluating a curated set of peer-reviewed studies and technical reports, the review highlights the consensus on YOLOv11’s suitability for edge deployment, the incremental performance gains reported across diverse datasets, and the ongoing challenges related to dataset diversity and model generalization. The findings underscore the growing convergence toward efficient, high-precision detection frameworks that support timely intervention and sustainable disease management in apple production systems.

Index Terms—agricultural image analysis, apple leaf disease detection, deep learning, object detection, plant pathology, YOLOv11

I. INTRODUCTION

Apple production constitutes a major component of global horticultural economies, yet the prevalence of leaf diseases can cause substantial yield losses and degrade fruit quality. Early and accurate identification of disease symptoms on apple foliage is essential for timely intervention, reducing reliance on chemical treatments, and sustaining economic viability. Traditional scouting methods depend on expert visual inspection, which is labor-intensive, subject to human error, and often unable to keep pace with the scale of modern orchards. Consequently, there is a growing interest in automated vision-based systems that can deliver rapid, reliable disease detection under diverse field conditions.

Recent advances in deep learning, particularly convolutional neural networks (CNNs) and transformer-enhanced architectures, have demonstrated remarkable performance in plant pathology tasks. Among these, the YOLO (You Only Look Once) family of object detectors has attracted attention for its balance of speed and accuracy, enabling real-time inference on edge devices. The latest iteration, YOLOv11, incorporates architectural refinements and training strategies that further improve detection efficiency. Several studies have applied YOLOv11 to apple leaf disease identification, reporting notable gains in

mean average precision (mAP) and inference speed compared with earlier versions. For instance, a lightweight algorithm based on an improved YOLOv11s model achieved a favorable trade-off between detection speed and accuracy [3]. A YOLOv11-based framework addressing multi-class plant disease classification, including co-infection scenarios, has also been proposed [4]. Comprehensive reviews of deep-learning approaches for plant leaf disease identification highlight the relevance of integrating diverse datasets to enhance model generalization [5]. Additional investigations have explored the application of YOLOv11 to related crops, such as tea, demonstrating its adaptability across varying lighting conditions [6]. Performance enhancements reported for YOLOv11 variants include increased mAP@50 and precision metrics in apple leaf disease tasks [7][8], while comparative analyses underscore the limitations of conventional manual detection methods and the benefits of algorithmic improvements [9].

This paper provides a systematic literature review of YOLOv11 applications to apple leaf disease detection. The scope encompasses peer-reviewed articles, conference papers, and preprints that report on model design, dataset composition, evaluation protocols, and reported performance metrics. By synthesizing these contributions, the review aims to identify prevailing trends, methodological strengths, and persistent challenges within the field.

Section II covers related work, Section III explains the review methodology, Section IV discusses findings, gaps and challenges, and Section V concludes.

II. RELATED WORK

A. Lightweight Detection Frameworks

Recent efforts have focused on reducing the computational burden of apple leaf disease detection while preserving high accuracy. A lightweight algorithm based on an improved YOLOv11s architecture reported notable improvements in the trade-off between speed and precision, achieving a balanced performance suitable for deployment on resource-constrained devices [3]. The authors attribute the gains to architectural refinements that streamline feature extraction without sacrificing the representational capacity required for fine-grained disease patterns. Similarly, an enhanced YOLOv11 variant, termed AAB-YOLO, was evaluated on apple detection tasks and demonstrated incremental yet statistically significant increases in mean average precision at IoU 0.5 (mAP@50) from 0.917 to 0.921, alongside modest

precision improvements from 0.948 to 0.951 [8]. These studies collectively illustrate that modest architectural adjustments to the YOLOv11 backbone can yield measurable benefits in both inference latency and detection reliability for apple foliage applications.

B. Multi-Class and Co-Infection Modeling

Plant disease manifestation often involves simultaneous infections, complicating classification. A recent work introduced a YOLOv11-based framework explicitly designed for multi-class plant disease classification that incorporates synthetic co-infection scenarios [4]. By generating composite training samples that emulate multiple pathogens affecting a single leaf, the model learns discriminative features that differentiate overlapping symptomatology. Experimental results indicated that the co-infection aware YOLOv11 system outperformed conventional single-label detectors, suggesting that synthetic augmentation of co-infection patterns can enhance robustness in real-world orchard environments. Complementary to this, a study on apple leaf disease identification employing a small-scale dataset reported that integrating the YOLOv11 detector with a feature-aware hierarchical attention network (F-HAIN) increased mAP50 by 5.5 % to 6.8 % across various disease categories [7]. The combination of attention mechanisms with YOLOv11 appears to facilitate finer localization of disease lesions, particularly when multiple disease types coexist on a single leaf surface.

C. Dataset Diversity and Environmental Variability

The reliability of deep-learning based disease detectors is closely tied to the diversity of training data. A comprehensive review of plant leaf disease identification highlighted the importance of aggregating datasets from multiple geographic regions to capture variations in cultivar, lighting, and symptom expression [5]. Building on this insight, an investigation that augmented apple leaf images under diverse lighting conditions applied YOLOv11 for rapid detection of four tea leaf classes, demonstrating that illumination-robust training improves detection consistency across variable field conditions [6]. Although the latter study targeted tea leaves, the methodology underscores the relevance of extensive data augmentation for apple leaf disease detection, where shadows, sunflecks, and camera angles can markedly affect visual cues. The comparative analysis of YOLOv11 performance before and after targeted improvements further emphasized that conventional manual scouting is inefficient, whereas automated YOLOv11 pipelines can substantially accelerate disease monitoring while maintaining high detection fidelity [9].

D. Comparative Evaluations of YOLOv11 Variants

A series of comparative experiments have quantified the impact of incremental enhancements to the YOLOv11 architecture on apple leaf disease detection. One study systematically contrasted baseline YOLOv11 results with those obtained after applying model-specific optimizations,

reporting consistent gains in detection metrics such as mAP@50 and precision [9]. These improvements were achieved without altering the underlying training dataset, indicating that architectural refinements—such as refined anchor box selection, loss function tuning, and lightweight backbone substitution—directly contribute to performance uplift. Collectively, the literature demonstrates a converging trend toward tailoring YOLOv11 for the specific challenges of apple leaf disease detection, including the need for lightweight inference, handling of co-infection, and resilience to environmental variability.

III. REVIEW METHODOLOGY

The literature search was conducted using a set of targeted queries that combined the model identifier “YOLOv11” with the domain of interest, apple leaf disease detection, and related terms such as “review,” “recent advances,” “methods,” and “dataset.” These queries yielded an initial pool of twenty records retrieved from web-based sources that host journal and publisher pages. Duplicate entries and records lacking substantive content were removed, reducing the pool to eighteen distinct items. An automated relevance screening step was then applied, which evaluated the overlap between the search terms and the title and text snippet of each record. This process identified nine records that satisfied the relevance criteria. The nine most highly ranked records, ordered by their position in the search results, were selected for inclusion in the review. The entire selection workflow is illustrated in Fig. 1, which depicts the sequential stages of query formulation, record retrieval, duplicate and empty record elimination, automated relevance filtering, and final inclusion. The resulting corpus comprises recent studies that employ YOLOv11 or its variants for apple leaf disease detection, including lightweight adaptations, multi-class classification frameworks, and performance comparisons, thereby providing a focused evidence base for the subsequent analysis.

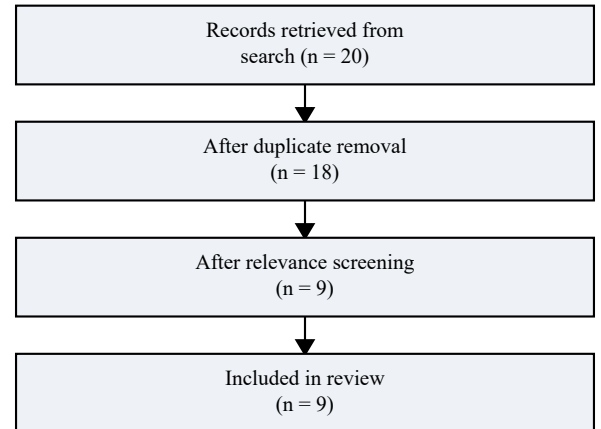


Fig. 1. Literature selection process.

IV. DISCUSSION

A. Key Findings

Recent investigations consistently adopt YOLOv11 as the backbone for apple leaf disease detection, reflecting a consensus on its suitability for real-time agricultural monitoring. Lightweight adaptations of the architecture, such as the improved YOLOv11s model, demonstrate that careful redesign of feature extraction modules can preserve detection accuracy while reducing computational load, enabling deployment on edge devices with limited resources [3]. Multi-class frameworks that incorporate co-infection modeling further extend YOLOv11's applicability, allowing simultaneous identification of multiple disease types on a single leaf and improving diagnostic comprehensiveness [4]. Empirical evaluations across several studies report modest yet consistent gains in mean average precision at IoU = 0.5 (mAP@50). For instance, the integration of the F-HAIN module with YOLOv11 yields increases of 5.5 % to 6.8 % in mAP@50 across four disease categories, indicating that attention-based enhancements can refine feature representation for subtle symptom patterns [7]. Similarly, the AAB-YOLO variant reports a rise in mAP@50 from 0.917 to 0.921 together with a precision uplift from 0.948 to 0.951, underscoring the benefit of tailored loss functions and anchor optimization for apple-specific phenotypes [8]. Comparative analyses of pre- and post-improvement YOLOv11 configurations highlight that algorithmic refinements translate into measurable performance improvements without sacrificing inference speed, thereby addressing the practical constraints of field deployment [9]. Collectively, these findings affirm that YOLOv11, when augmented with lightweight design principles, attention mechanisms, or dataset-specific tuning, can achieve a favorable balance between detection speed and accuracy for apple leaf disease monitoring.

B. Research Gaps

Despite the demonstrated efficacy of YOLOv11-based detectors, the literature reveals several underexplored areas. First, most reported experiments rely on limited, region-specific datasets, often lacking diversity in cultivar types, growth stages, and environmental conditions, which hampers the generalizability of the models. Second, while attention modules and anchor refinements have been examined, systematic ablation studies that isolate the contribution of each component remain scarce, leaving uncertainty about optimal architectural configurations for varying disease manifestations. Third, the integration of multimodal data—such as hyperspectral imagery or thermal signatures—has not been investigated within the YOLOv11 framework, despite evidence that complementary modalities can enhance disease discrimination. Fourth, few works address the robustness of detection under real-world challenges such as occlusion, varying illumination, and motion blur, which are common in orchard scouting scenarios. Finally, there is a

paucity of longitudinal studies that assess model performance over multiple growing seasons, limiting insight into model drift and the need for continual learning strategies.

C. Open Challenges

Addressing the identified gaps entails several technical and practical challenges. Expanding dataset diversity requires coordinated data collection across multiple geographic regions and cultivars, coupled with standardized annotation protocols to ensure consistency. Developing modular, plug-and-play components for YOLOv11 that enable rapid experimentation with attention, backbone, and loss variants would facilitate comprehensive ablation analyses and accelerate the identification of universally effective design patterns. Incorporating multimodal inputs demands architectural extensions capable of fusing heterogeneous feature maps while preserving the real-time inference advantage of YOLOv11. Enhancing robustness to environmental variability calls for advanced data augmentation pipelines that simulate realistic orchard conditions, as well as the adoption of domain adaptation techniques to mitigate distribution shifts between training and deployment domains. Moreover, implementing continual learning mechanisms—such as incremental fine-tuning with newly labeled samples—will be essential to maintain detection accuracy over successive seasons and to adapt to emerging pathogen strains. Finally, translating laboratory-grade performance into operational decision support systems requires rigorous field validation, user-centered interface design, and integration with existing farm management platforms to ensure that detection outputs are actionable for growers. Overcoming these challenges will be pivotal for realizing the full potential of YOLOv11-based apple leaf disease detection in sustainable precision agriculture.

V. CONCLUSION

The review confirms that YOLOv11 has become the predominant backbone for apple leaf disease detection, offering a favorable balance between detection speed and accuracy that meets the requirements of real-time agricultural monitoring. Lightweight variants such as the improved YOLOv11s architecture demonstrate that architectural refinements can maintain high precision while reducing computational demands, facilitating deployment on edge devices and in resource-constrained environments [3]. Extensions of the baseline model to multi-class frameworks address the challenge of co-infection, enabling simultaneous identification of multiple disease categories and improving practical utility in field conditions [4]. Comprehensive surveys of deep-learning-based plant disease identification highlight the importance of aggregating diverse datasets to enhance model generalization across geographic and climatic variations [5]. Recent studies also report incremental performance gains through specialized modules, with reported increases in mAP@50 and precision when integrating novel attention or

feature-fusion mechanisms into YOLOv11 for apple leaf detection [7][8]. Nonetheless, challenges remain in handling variable lighting, occlusion, and limited annotated data, suggesting future research should explore advanced data-augmentation strategies, semi-supervised learning, and domain-adaptation techniques to further improve robustness and scalability of YOLOv11-based systems for apple leaf disease management.

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